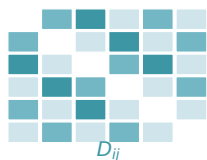


(A) From coordinates to one canonical record

Invariant geometric evidence

M-donor distances + donor-donor distances



refined donor signatures

donor attribute: element
geometric $\vec{f}_i = [d_{M,i}, \text{sort}(D_{i*})]$

N · 1.86 · 2.63 · 2.81 ...

N · 1.86 · 2.71 · 2.79 ...

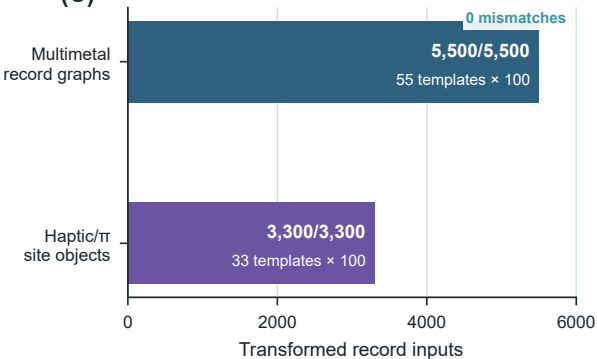
O · 1.99 · 2.63 · 2.71 ...

lexicographic order

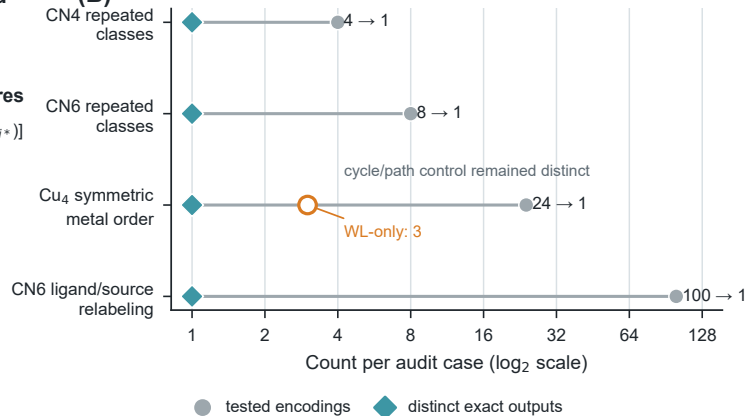
L1 L2 L3 L4 L5 L6

remaining exact ties
→ minimize the full record

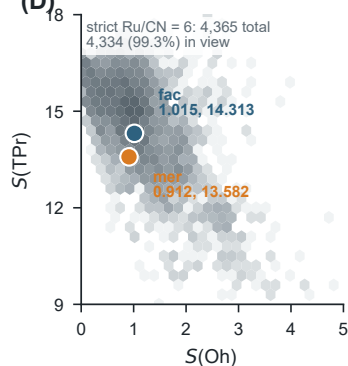
(C) Expanded-scope exact recovery



(B) Nuisance encodings converge to one exact State



(D) Corrected CShM atlas



Trans relations

